

Local Qwen Audio Assistant

TASK-3

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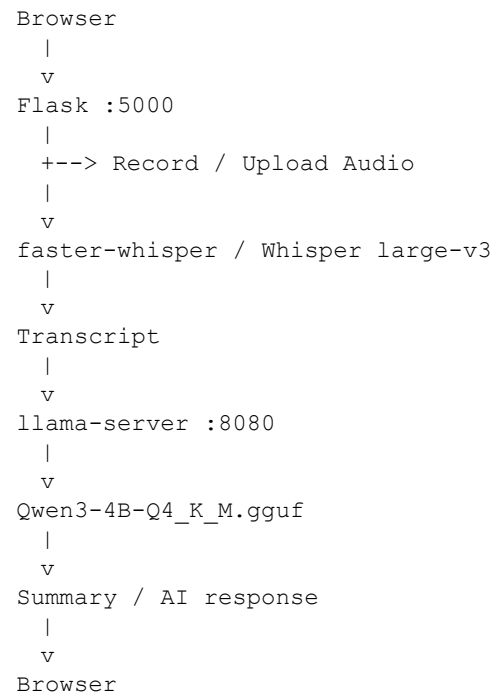
Detailed Technical Project Report

Local Windows audio transcription and summarization pipeline using Flask, faster-whisper, Whisper large-v3, Qwen3-4B and llama.cpp.

1. Executive Summary

The application accepts audio through direct browser recording or existing-file upload. Flask orchestrates the workflow. faster-whisper runs Whisper large-v3 locally for speech-to-text. The transcript is then sent through HTTP to llama.cpp's local server, which runs the Qwen3-4B GGUF model for summarization and other language tasks.

2. Architecture



3. Models and Software

Component	What it does	Why needed
Flask	Web application and orchestration	Connects browser, transcription and Qwen
faster-whisper	Runs Whisper transcription	Converts audio to text efficiently
Whisper large-v3	Speech recognition model	Understands spoken audio
Qwen3-4B	Local language model	Summarizes and processes transcript text
llama.cpp	GGUF inference runtime	Runs Qwen locally
llama-server	HTTP server for llama.cpp	Provides the Qwen API on port 8080
requests	Python HTTP client	Sends transcript prompts to Qwen

4. End-to-End Workflow

Audio input

The user records audio in the browser or uploads an existing recording.

Transcription

Flask passes the audio to faster-whisper, which uses Whisper large-v3 to produce text.

Prompt

The transcript becomes the input to the language model.

Inference

Flask sends an HTTP request to llama-server at 127.0.0.1:8080/v1/chat/completions.

Generation

Qwen3-4B generates the summary or requested language output.

Display

Flask returns the transcription and generated result to the browser.

5. Why Two AI Models?

Whisper and Qwen have different responsibilities. Whisper specializes in speech recognition, while Qwen specializes in language understanding and generation.

```
Speech -> Whisper -> Text -> Qwen -> Summary
```

6. Python Requirements

```
Flask>=3.1,<3.2
requests>=2.34,
<3
faster-whisper>
=1.2,<2
```

These are Python dependencies. The Qwen GGUF file, Whisper model assets and llama.cpp executable are separate local dependencies.

7. Verification

The environment verified Python 3.12.10, faster-whisper, Flask and requests. The Qwen health endpoint returned status ok. A Python requests session with trust_env disabled returned HTTP 200 from the Qwen chat endpoint. The Flask application successfully started and reported that Whisper was ready.

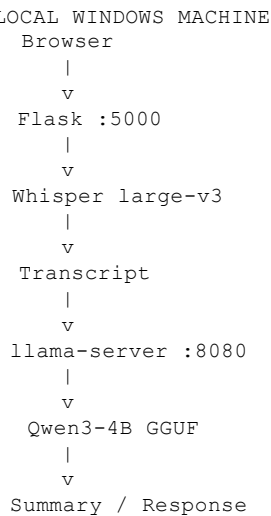
8. Troubleshooting

Issues	Resolution
Qwen health fails	Start llama-server and verify port 8080.
Python HTTP 502	Use requests.Session() and set session.trust_env = False.
Flask missing	Install Flask into E:\AI\env.
requests missing	Install requests into E:\AI\env.
Whisper missing	Verify faster-whisper in E:\AI\env and the model cache.
Command syntax errors	PowerShell uses backtick continuation; CMD does not so ensure use power shell.

9. Final Technical Description

This project is a local multi-model AI audio assistant built with Flask, faster-whisper, Whisper large-v3, Qwen3-4B and llama.cpp. It supports both browser recording and existing audio-file uploads. faster-whisper performs local speech-to-text transcription, while Qwen3-4B performs natural-language processing and summarization through the llama.cpp OpenAI-compatible HTTP server. Flask orchestrates the complete workflow and provides the browser interface.

10. Architecture at a Glance



11. Scope

The scope of this project is to develop a fully local AI-powered audio assistant that can take spoken conversations or existing audio recordings and automatically convert them into useful written information. The system is designed to run on a Windows machine without depending on cloud-based AI services for its core processing.

The application provides two ways to provide audio: the user can either record audio directly through the web interface or upload an existing audio file. Once the audio is received, the system uses Whisper through faster-whisper to convert the speech into text. The generated transcript is then passed to the locally running Qwen3-4B language model, which understands the transcript and can generate a concise summary or other requested text-based output.

The project uses Flask as the web application layer, while llama.cpp/llama-server provides an HTTP interface for running the Qwen model locally. This creates a complete pipeline where the browser, speech recognition model, and language model work together.

12. OUTPUT

Local AI Audio Assistant

Record or upload audio → local Whisper transcription → local Qwen summary

1. Choose audio

 Start Recording

 Upload Audio

 Clear

00:00

No audio selected.

Microphone error: Requested device not found

2. Process audio

 Transcribe & Summarize

Transcript

Nothing transcribed yet.

Summary

Nothing summarized yet.

Local AI Audio Assistant

Record or upload audio → local Whisper transcription → local Qwen summary

1. Choose audio

 Start Recording

 Upload Audio

 Clear

00:00

 Englishsample.flac (987 KB)

 Complete. Audio was processed locally.

2. Process audio

 Transcribe & Summarize

Transcript

Hi there, this is a sample voice recording created for speech synthesis testing. The quit brown fox jumps over the lazy dog. Just a fun way to include every letter of the alphabet. Numbers like 1, 2, 3 are spoken clearly. Let's see how well this voice captures tone, timing, and natural rhythm. This audio is provided by samplefiles.com.

Summary

SUMMARY:

The transcript is a sample voice recording created for testing speech synthesis. It includes a well-known pangram that uses every letter of the alphabet and features numbers for clarity. The purpose is to evaluate the voice's ability to capture tone, timing, and natural rhythm.

2. Process audio

 Transcribe & Summarize

Transcript

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Summary

SUMMARY:

The transcript is a sample voice recording created for testing speech synthesis. It includes a well-known pangram that uses every letter of the alphabet and features numbers for clarity. The purpose is to evaluate the voice's ability to capture tone, timing, and natural rhythm.

KEY POINTS:

- The recording is a test for speech synthesis.
- It includes a pangram using all letters of the alphabet.
- Numbers 1, 2, 3 are spoken clearly.

ACTION ITEMS:

- None

IMPORTANT DETAILS:

- Website: samplefiles.com
- Pangram: "The quit brown fox jumps over the lazy dog"
- Numbers: 1, 2, 3
- Purpose: Testing tone, timing, and natural rhythm of the voice.